PERGAMON

Energy Conversion and Management 42 (2001) 529–538

ENERGY CONVERSION & MANAGEMENT

www.elsevier.com/locate/enconman

# The potential of using vegetable oil fuels as fuel for diesel engines

Recep Altın [a], Selim Çetinkaya [b,*], Hüseyin Serdar Yücesu [c]

[a] *Projects Coordination Unit, Ministry of Education, Ankara 06500, Turkey*
[b] *Technical Education Faculty, Gazi University, Ankara 06500, Turkey*
[c] *Karabük Technical Education Faculty, Karaelmas University, Karabük 78100, Turkey*

Received 9 December 1999; accepted 24 August 2000

**Abstract**

Vegetable oils are produced from numerous oil seed crops. While all vegetable oils have high energy content, most require some processing to assure safe use in internal combustion engines. Some of these oils already have been evaluated as substitutes for diesel fuels. The effects of vegetable oil fuels and their methyl esters (raw sunflower oil, raw cottonseed oil, raw soybean oil and their methyl esters, refined corn oil, distilled opium poppy oil and refined rapeseed oil) on a direct injected, four stroke, single cylinder diesel engine performance and exhaust emissions was investigated in this paper. The results show that from the performance viewpoint, both vegetable oils and their esters are promising alternatives as fuel for diesel engines. Because of their high viscosity, drying with time and thickening in cold conditions, vegetable oil fuels still have problems, such as flow, atomization and heavy particulate emissions. © 2000 Elsevier Science Ltd. All rights reserved.

*Keywords:* Vegetable oil fuels; Alternative fuels; Biodiesel

## 1. Introduction

Recent petroleum crises, rapidly increasing prices and uncertainties concerning petroleum availability let the scientists work on alternative fuel sources, so vegetable fuel studies become current among various investigations. The idea of using vegetable oils as fuel for diesel engines is not new. Rudolph diesel used peanut oil to fuel one of his engines at the Paris Exposition of 1900.

* Corresponding author. Fax: +90-312-212-0059.
*E-mail address:* scetin@tef.gazi-edu.tr (S. Çetinkaya).


PII: S0196-8904(00)00080-7

Blumberg and Ford [1], made short and long term (200 h) engine performance and emission tests using eight different fuel samples: 2D diesel fuel; 30% cottonseed oil, 70% 2D diesel fuel (by volume); 50% cottonseed oil, 50% 2D diesel fuel; 65% cottonseed oil, 35% 2D diesel fuel; 80% cottonseed oil, 20% 2D diesel fuel; 50% cottonseed oil, 50% transesterified cottonseed oil; 50% transesterified cottonseed oil; 50% 2D diesel fuel; and 100% cottonseed methyl ester. They reported that short term results have been more desirable than long term results. Long term tests showed carbon deposits, ash and wear in the combustion chamber and sticky gum content in fuel line elements [1].

Schinstock et al. [2] made 200 h engine performance tests using 25/75 blends (volumetric) of soybean and sunflower oil fuel with diesel fuel. They reported that engine torque values with mixtures have been greater than with pure diesel operation [2].

Hemmerlein et al. [3] reported on rapeseed oil as a fuel and concluded that the physical and chemical properties of rapeseed oil as a fuel are very similar to those of diesel fuel, and in a long term basis, it can be used in diesel engines. They also reported comparative test results of six different engines fueled with rapeseed oil and diesel fuel. With rapeseed oil fuel

- torque, power and $NO_x$ emissions of 5 engines out of 6 were better;
- HC emissions of 5 engines out of 6 were worse;
- CO emission values were worse in all engines;
- two of the engines showed better endurance test results.

Energy consumption values of all the engines were about the same for both fuels [3].

Schumacher [4] reported on soybean methyl ester as a fuel in a Dodge truck and concluded that it can be used in diesel engines with little or no difficulty. He also reported that when using 10–20–30–40–50% (volumetric) soybean methyl ester–diesel fuel mixtures, as the soybean methyl ester ratio in the mixture increased, the power, smoke intensity, CO and HC emissions decreased, $NO_x$ emissions and fuel consumption increased [4].

In general, it has been reported by most researchers that if raw vegetable oils are used as diesel engine fuel, engine performance decreases, CO and HC emissions increase and $NO_x$ emissions also decrease accordingly [5–9]. Results show that vegetable oil fuels have the potential of supplying some of the energy needed by agricultural areas. The most important advantage of vegetable oils is that they are renewable energy sources compared to the limited resources of petroleum. Vegetable oils show promise of providing all the liquid fuel needed on a typical farm by diverting 10% or less of the total acreage to fuel production [10]. The extraction and procession of vegetable oils are simple low energy processes that make use of equipment not unlike that with which farmers are already familiar.

Among the vegetable oil seeds that can be grown as domestic field crops, cottonseed and sunflower seed are the major productions of Turkey (Table 1) [11]. However, as seen from the table, crop productions are inconsistent according to harvest area, climatic conditions, etc. Turkey is one of the major sunflower producers of the world. By the end of 1988, it had 6.18% of the total world production areas [10]. Refined sunflower oil and refined cottonseed oil productions in 1997 were 514 635 tons and 20 546 tons, respectively [11].

The objectives of this study were to evaluate comparatively the performance and exhaust emissions of a diesel engine using 100% refined vegetable oil and their methyl esters. Because of its high viscosity, vegetable oils were heated before the fuel pump and before the injectors to minimize its resistance to flow.

Table 1
Production levels of some oil seeds of Turkey [11]

| Crops | Production (ton) | | |
|---|---|---|---|
| | 1990 | 1995 | 1997 |
| Sunflower | 860 000 | 900 000 | 900 000 |
| Rapeseed | 2100 | 9 | 10 |
| Cottonseed | 1 047 360 | 1 262 583 | 1 193 286 |
| Soybean | 162 000 | 75 000 | 40 000 |
| Sesame | 39 000 | 30 000 | 28 000 |
| Opium seed | 5153 | 28 249 | 10 948 |

## 2. Experiment and test procedure

Tests were conducted at the Technical Education Faculty, Automotive Laboratory of Gazi University. A single cylinder, four stroke, direct injected diesel engine was selected for the tests. Technical specifications of the test engine are summarized in Table 2. Tests were held on a laboratory test bench which consisted of an electrical dynamometer, a cooling tower, an exhaust gas aspiration system and engine mounting elements, as shown in Fig. 1. The engine fuel system was modified by adding two fuel meters and one 3 way, hand operated control valve which allowed rapid switching between the standard diesel fuel and the test fuel. The fuel return line of the engine was connected to the fuel meter. Air consumption was measured with a Go-Power M-5000 flow meter, and an exhaust emission values were measured with a Gaco–Sn gas analyzer and an Okuda Koki smoke meter.

The aim of the study was to investigate the effect of vegetable based oils on compression engine performance and emissions. Diesel fuel and nine different vegetable oil fuels (raw sunflower oil, raw cottonseed oil, raw soybean oil and their methyl esters, refined corn oil, distilled opium poppy oil and refined rapeseed oil) were selected and used as test fuels. The raw oil fuels were purchased from oil plants, refined oil fuel was purchased from the commercial market and methyl esters were

Table 2
Technical specifications of the test engine [12]

| | |
|---|---|
| Type | Superstar, 7710 |
| Year | 1980 |
| Number of cylinders | 1 |
| Compression ratio | 17:1 |
| Cylinder diameter | 98 mm |
| Stroke | 100 mm |
| Clearance volume | 770 $cm^3$ |
| Maximum speed | 2000 1/min |
| Power | 7–8 HP at 1500 1/min |
| Type of cooling | Water |
| Injection timing | 27° BTDC |
| Starting | By hand |
| Injector opening pressure | 175 $kg/cm^2$ |

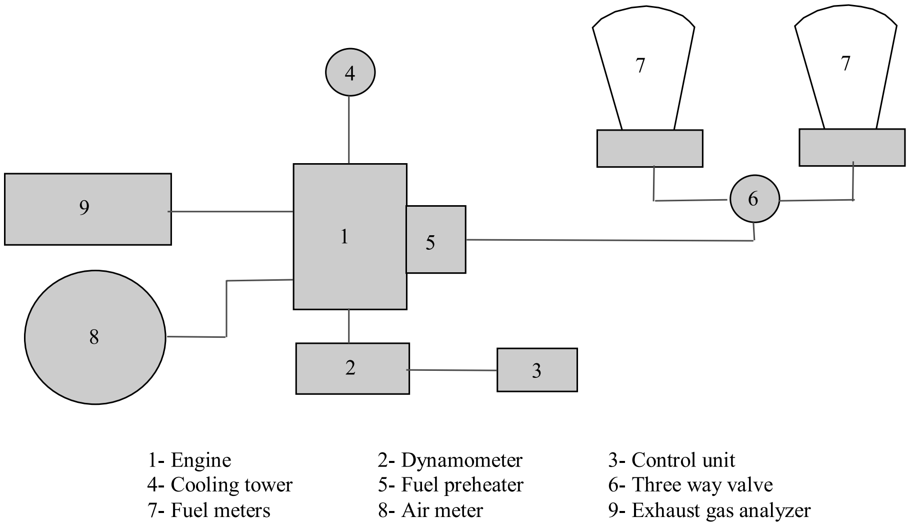


Fig. 1. Schematic layout of the test system.

produced in the laboratory environment. Physical and chemical specifications of the vegetable oil fuels used are summarized in Table 3.

Because of their higher viscosity, raw vegetable oils were heated before the fuel pump and before the injectors to minimize their resistance to flow. Two thermostatically controlled electrical heaters were used for this purpose, and the temperatures were about 80°C for both regions.

The parameters were obtained from full load and variable speed performance tests. The engine was operated on diesel fuel first and then on vegetable oil and methyl esters. After the raw vegetable oil tests, the engine was run about 20 min at idle speed to replace the test fuel in the system

Table 3
Physical and chemical specifications of the vegetable oil fuels used

| Fuel type | Calorific value (kJ/kg) | Density (g/dm$^3$) | Viscosity (mm$^2$/s) | | Cetane number | Flame point (°C) | Chemical formula |
|---|---|---|---|---|---|---|---|
| | | | 27°C | 75°C | | | |
| Diesel fuel | 43 350 | 815 | 4.3 | 1.5 | 47[a] | 58 | $C_{16}H_{34}$ |
| Raw sunflower oil | 39 525 | 918 | 58 | 15 | 37.1[a] | 220 | $C_{57}H_{103}O_6$ |
| Sunflower methyl ester | 40 579 | 878 | 10 | 7.5 | 45–52 | 85 | $C_{55}H_{105}O_6$ |
| Raw cottonseed oil | 39 648 | 912 | 50 | 16 | 48.1[a] | 210 | $C_{55}H_{102}O_6$ |
| Cottonseed methyl ester | 40 580 | 874 | 11 | 7.2 | 45–52 | 70 | $C_{54}H_{101}O_6$ |
| Raw soybean oil | 39 623 | 914 | 65 | 9 | 37.9[a] | 230 | $C_{56}H_{102}O_6$ |
| Soybean methyl ester | 39 760 | 872 | 11 | 4.3 | 37 | 69 | $C_{53}H_{101}O_6$ |
| Corn oil | 37 825 | 915 | 46 | 10.5 | 37.6[a] | 270–295 | $C_{56}H_{103}O_6$ |
| Opium poppy oil[a] | 38 920 | 921 | 56 | 13 | – | – | $C_{57}H_{103}O_6$ |
| Rapeseed oil[b] | 37 620 | 914 | 39.5 | 10.5 | 37.6[a] | 275–290 | $C_{57}H_{105}O_6$ |

[a] Values for opium poppy oil were taken from Doysan Ltd. [13].
[b] Values for rapeseed oil were taken from Paksoy Ltd. [14].

with diesel fuel. The test matrix consisted of 10 speeds ranging from 900 to 1800 1/min with 100 1/min steps for each fuel operation. All engine performance data were corrected to the standard atmospheric conditions given below.

The power correction factor used was (according to Turkish Standards, TS 1231) [15]:

$$K_{\mathrm{d}} = f_{\mathrm{a}}^{f_{\mathrm{m}}},$$

where

$$f_{\mathrm{a}} = \left(\frac{99}{P_{\mathrm{a}}}\right)\left(\frac{T_{\mathrm{a}}}{298}\right)^{0.7},$$

$$f_{\mathrm{m}} = 0.036 q_{\mathrm{c}} - 1.14,$$

$$q_{\mathrm{c}} = q/r.$$

$T_{\mathrm{a}}$ is ambient temperature (K), $P_{\mathrm{a}}$ is ambient pressure (kPa), $q$ is fuel delivery rate (mg/cycle) and $r$ is compressor pressure ratio (for naturally aspirated engines, $r = 1$).

## 3. Results and discussion

### 3.1. Performance

The variations of maximum engine torque values in relation with the fuel types are shown in Fig. 2. The maximum torque with diesel fuel operation was 43.1 Nm at 1300 1/min. For ease of comparison, this torque was assumed 100% as the reference. The observed maximum torque values of the vegetable oil fuel operations were also at about 1300 1/min but less than the diesel fuel value for each fuel. The maximum torque differences between the reference value and the peak values of the vegetable oil fuels were about 10% obtained with raw sunflower oil, raw soybean oil

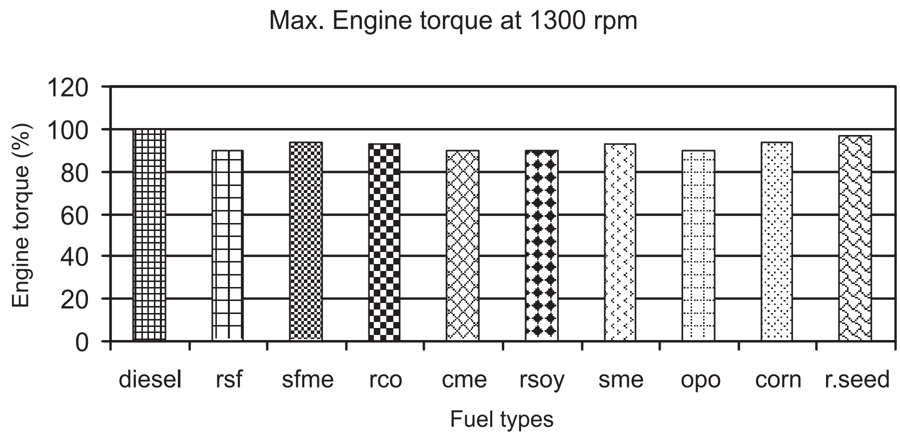


Fig. 2. The variation of engine torque in relation with the fuel types.

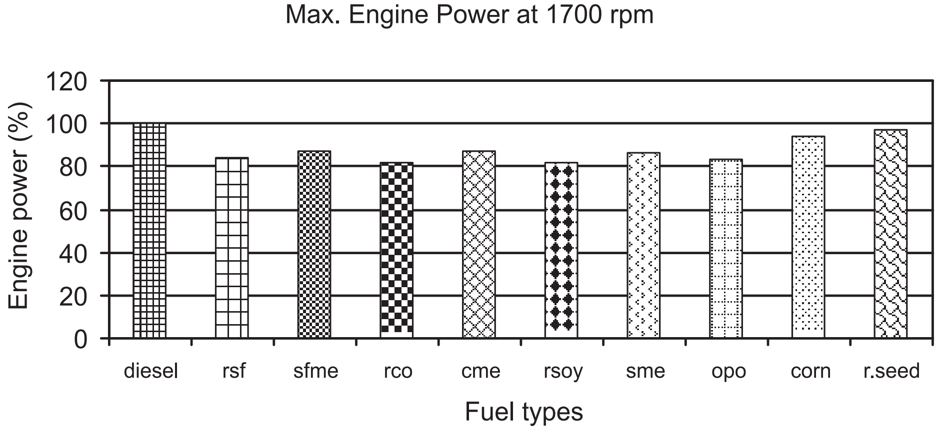


Fig. 3. The variation of engine power in relation with the fuel types.

and opium poppy oil fuels. The minimum torque difference was about 3% between the reference value and the peak values of refined corn oil and rapeseed oil fuels. These results may be due to the higher viscosity and lower heating values of vegetable oils.

The variations of maximum engine power values in relation with the fuel types are shown in Fig. 3. The maximum power with diesel fuel operation was 7.45 kW at 1700 1/min. As before, this power was assumed 100% as the reference. Observed maximum power values of the vegetable oil fuel operations were also at about 1700 1/min but less than the diesel fuel value for each fuel. The maximum power differences between the reference value and the peak values of the vegetable oil fuels were about 18%, obtained with raw cottonseed oil and raw soybean oil fuels. The minimum power difference was about 3% between the reference value and the maximum value of rapeseed oil fuels. These results may also be due to the higher viscosity and lower heating values of vegetable oils.

Specific fuel consumption is one of the important parameters of an engine and is defined as the consumption per unit of power in a unit of time. Test results showed that the obtained minimum specific fuel consumption values were in the vicinity of the maximum torque area. As shown in Fig. 4, the minimum specific fuel consumption values were 245 g/kW h with diesel fuel, 290 g/kW h with raw sunflower oil and 289 g/kW h with opium poppy oil at 1300 1/min. Specific fuel consumption values of the methyl esters were generally less than those of the raw oil fuels. The higher specific fuel consumption values in the case of vegetable oils are due to their lower energy content.

### 3.2. Exhaust emissions

The variations of CO emissions in relation with the fuel types are shown in Fig. 5. The minimum CO emission was about 2225 ppm with diesel fuel. On the other hand, maximum CO emissions were about 4000 ppm with rapeseed oil fuel, 3850 ppm with raw sunflower oil and 3800 ppm with corn oil fuel. Comparing raw and refined vegetable oils, relatively lower CO emissions were obtained with the esters. This is an expected result of better spraying qualities and more uniform mixture preparation of these fuels in this working range.

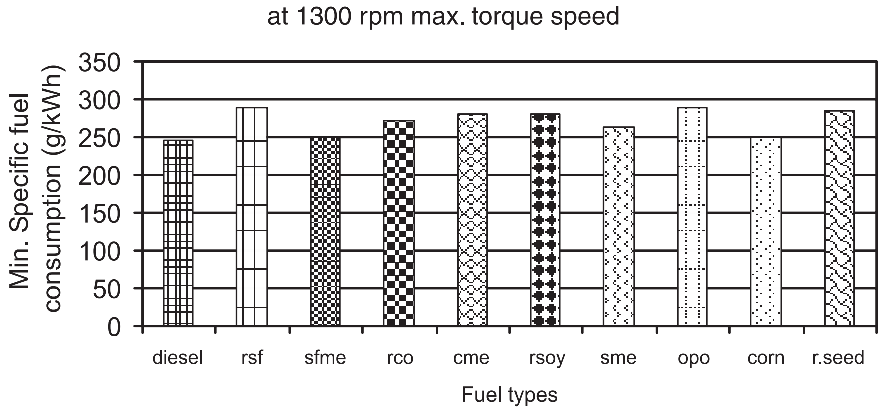


Fig. 4. The variation of minimum specific fuel consumption in relation with the fuel types.

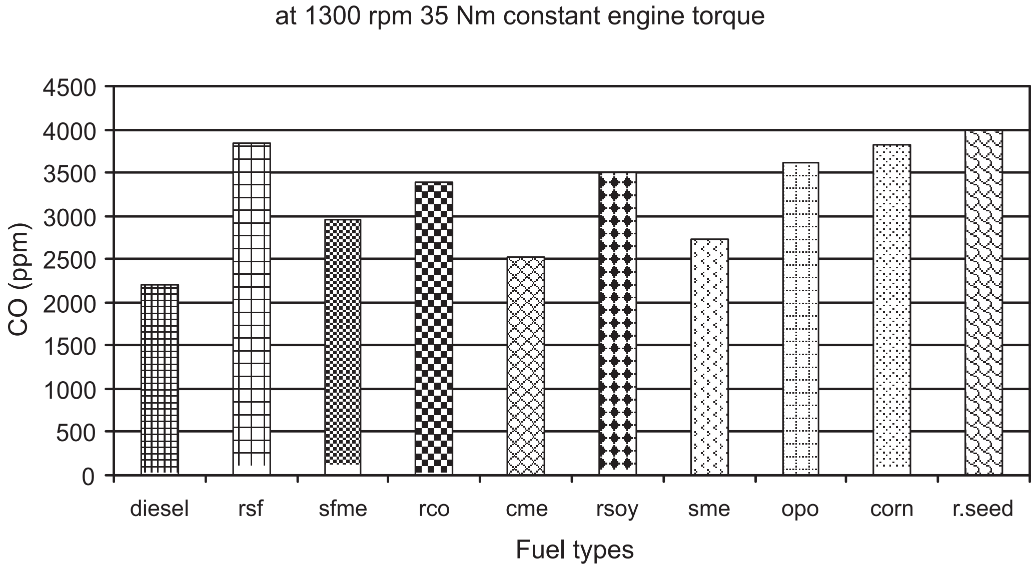


Fig. 5. The variation of CO emissions in relation with the fuel types.

The variations of $CO_2$ emissions in relation with the fuel types are shown in Fig. 6. Maximum $CO_2$ emissions were about 10.5% with diesel fuel, 10.25% with raw sunflower oil, 10.4% with sunflower methyl ester and 10.2% with raw soybean oil fuel. This is again an expected result of relatively better spraying qualities and more uniform mixture preparation of these fuels in this working range.

$NO_x$ emissions are usually a result in the higher combustion temperatures. The variations of $NO_2$ emissions in relation with the fuel used during 35 Nm torque load at 1300 1/min are shown in Fig. 7. Maximum $NO_2$ emission was 2100 mg/m$^3$ with diesel fuel. As seen in the graphics, $NO_2$ emissions with vegetable oil fuels were lower than those with diesel fuel, and the $NO_2$ values of the

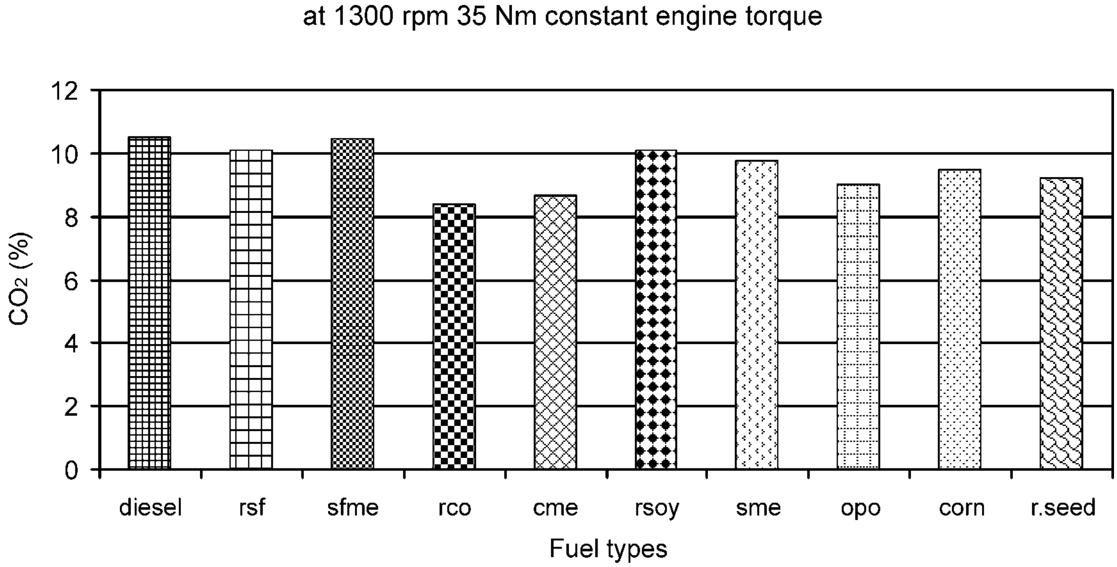


Fig. 6. The variation of $CO_2$ emissions in relation with the engine speed.

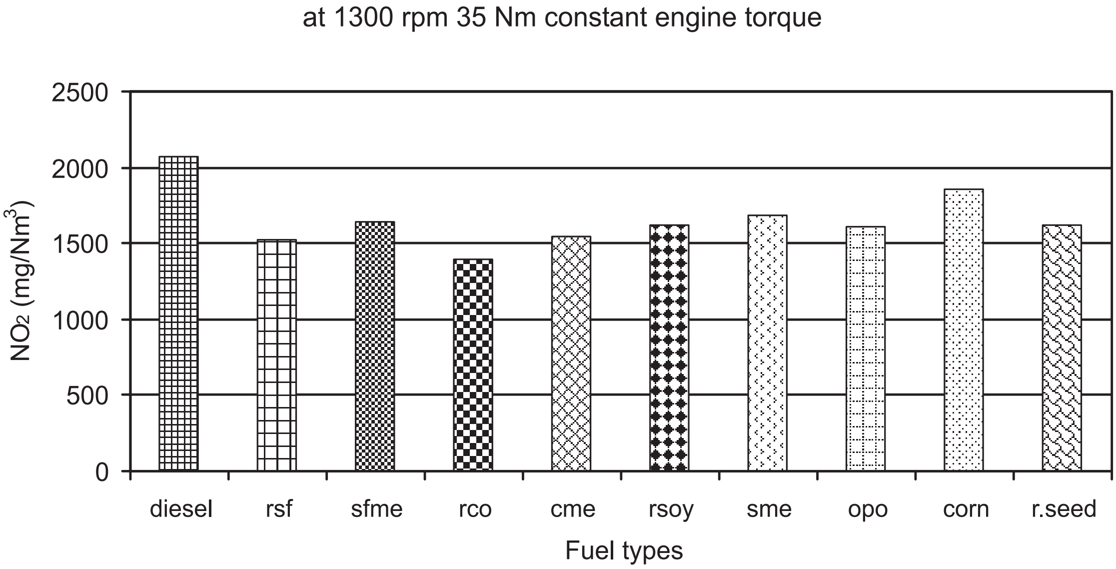


Fig. 7. The variation of $NO_2$ emissions in relation with the fuel types.

methyl esters were higher than those of the raw oil fuels. The most significant factor that causes $NO_2$ formation is the maximum combustion temperatures. Since injection particles of the vegetable oils were greater than those of diesel fuel, the combustion efficiency and maximum combustion temperatures with each of the vegetable oils were lower and $NO_2$ emissions were less.

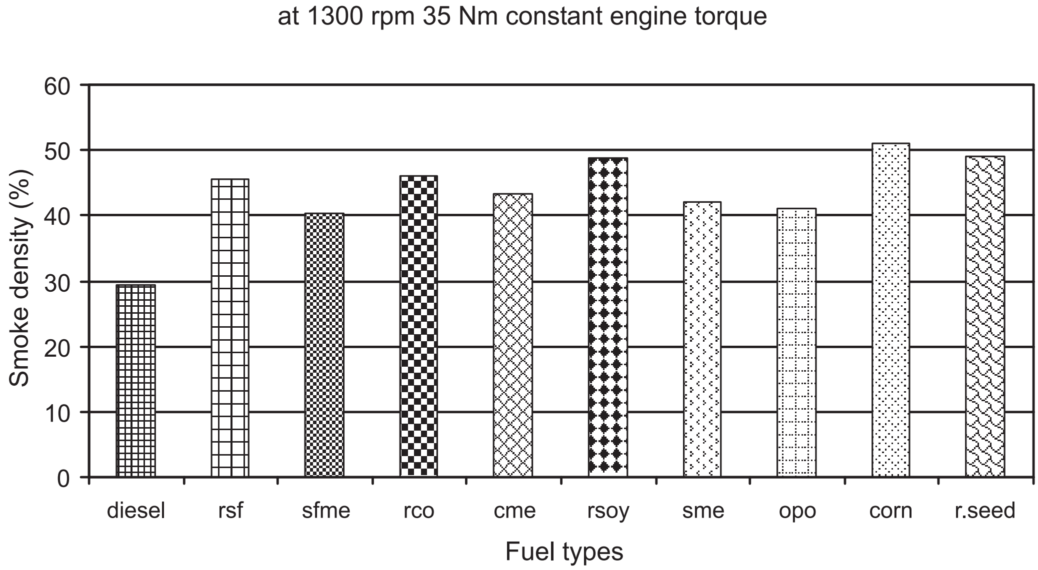


Fig. 8. The variation of particulate emissions in relation with the fuel types.

The variations of exhaust smoke opacity in percent in relation with the fuels used during 35 Nm torque load at 1300 1/min are shown in Fig. 8. Smoke opacity percentages during each of the vegetable oil operations were greater than that of diesel fuel. The minimum smoke opacity was 29.3% with diesel fuel, and the maximum smoke opacity values were 49% with soybean oil fuel and rapeseed oil fuel and 51% with corn oil fuel. The opacity values of the methyl esters were between those of diesel fuel and raw oil fuels. The greater smoke opacity percentages of vegetable oil fuels are mainly due to the heavier molecules of hydrocarbons.

## 4. Conclusions

In this study, it was concluded that

(i) Compared to diesel fuel, a little amount of power loss happened with vegetable oil fuel operations.
(ii) Particulate emissions of vegetable oil fuels were higher than that of diesel fuel, but on the other hand, $NO_2$ emissions were less.
(iii) Vegetable oil methyl esters gave performance and emission characteristics closer to the diesel fuel. So, they seem to be more acceptable substitutes for diesel fuel.
(iv) Raw vegetable oils can be used as fuel in diesel engines with some modifications.

Before starting wide application, there are some improvements that should be done, such as

(i) Fuel systems should be optimized for vegetable oil operation.
(ii) Vegetable oils are mainly consumed on the food market. Therefore, it has some unfavorable properties as fuel, such as high density, drying with time and gumming, lower cetane number,

heavier exhaust smoke etc. Fuel characteristics of vegetable oils should be improved.
(iii) At present, vegetable oils are more expensive than diesel fuel, and their annual harvesting values are not stable. As vegetable oil fuel consumption increases, production yield can be increased relatively, and the cost can be decreased by more mechanized farming.

## References

[1] Fort EF, Blumberg PN. Performance and durability of a turbocharged diesel fueled with cottonseed oil blends. Vegetable Oil Fuels Proceedings of the International Conference on Plant and Vegetable Oils as Fuel, ASAE, 1982.
[2] Schlick ML, Hanna MA, Schinstock JL. Soybean and sunflower oil performance in diesel engine. ASAE 1988;31:5.
[3] Hemmerlein M, Korte V, Richter HS. Performance, exhaust emission and durability of modern diesel engines running on rapeseed oil. SAE Paper, 910848, 1991.
[4] Schumacher L. Fueling used diesel engines with methyl-esters soybean oil. Soybean Research Documents Online, http://stratsoy.ag.uiuc.edu. /~stratsoy/research/usb9.html, internet, 1999.
[5] Sinha S, Misra NC. Diesel fuel alternative from vegetable oils. Exhibition and Communication Enterprises (India) Pvt Ltd, 1997.
[6] Goering GE, et al. Fuel properties of eleven vegetable oils. Trans of ASEA, 1982.
[7] Çetinkaya S. Ayçiçek Yağının Dizel Motoru Yakıtı Olarak Kullanilabilme Potansiyeli. Gazi Ü Teknik Eğitim Fakültesi, Doçentlik Takdim Tezi, Ankara, 1994.
[8] Altın R. An experimental investigation on use of vegetable oils as diesel engine fuels. PhD Thesis, Gazi University, Institute of Science and Technology, 1998.
[9] Shay EG. Diesel fuel from vegetable oils, status and opportunities. Natl Acad of Sciences, Washington DC, USA, 1993.
[10] Çetinkaya S. The potential of using sunflower oil as a fuel for diesel engines. Isı Bilimi ve Tekniği Dergisi (J Thermal Sci Technol), Cilt 17 Sayi: 4, Ankara, Haziran 1995.
[11] Türkiye İstatistik Yıllığı, 1998. TC Başbakanlìk DİE Yayın No: 2240. O.İ.E. Yayınları, Ankara, Mayıs 1999.
[12] Superstar diesel engine manual, 1980.
[13] Biochemical specifications of vegetable oils. Doysan Yağ Tic. A.Ş. 1997.
[14] Technical document of rapeseed oil. Paksoy AT, 1999.
[15] İçten Yanmalı Motorlar-Muayene ve Deney Esasları, TS 1231. TSE, (Institute of Turkish Standards), Ankara, Turkey, Nisan, 1991.